

Balloon Pop Lab - Conversation Log

LLM used: ChatGPT

Project: Balloon Pop (Pygame)

Purpose: Debug the starter code and implement Tasks 1–4 from README.md.

USER

I forked the repo, cloned it locally, played the game for 10 seconds, and shared the README and repository structure. Please follow the steps in the lab guide.

CHATGPT

Reviewed the lab guide and README. Identified the four required tasks: fix click detection, add balloon types, add a lives/miss system, and add a timed round. Confirmed the assignment requires separate commits for each task and before/after videos.

USER

Task 1.

CHATGPT

Explained the squared-distance bug in `game/click_detection.py` and provided the minimal fix: compare `distance_squared` against `balloon.radius * balloon.radius`. Suggested testing clicks near the balloon edge and committing the change separately.

USER

Task 2.

CHATGPT

Implemented the three balloon types concept: normal (+10), bonus (+25), and penalty (-10), each with a distinct color. Updated balloon spawning to randomly select a type and changed scoring to use each balloon's points.

USER

How do I commit both tasks?

CHATGPT

Explained that Task 1 and Task 2 must remain separate commits. The recommended commits were 'Task-1:Fix balloon click detection' and 'Task-2:Add different balloon types'.

USER

I pushed and showed the Git log.

CHATGPT

Verified the Git history showed separate Task 1 and Task 2 commits, with origin/main at the same commit as the local main branch.

USER

Task 3.

CHATGPT

Implemented the lives/miss system: 3 starting lives, losing one life when any balloon reaches the bottom, displaying remaining lives, and setting `game_over` when lives reach zero. Confirmed that popping a balloon does not reduce lives.

USER

Shared the current `update()` implementation.

CHATGPT

Adapted the Task 3 changes to the user's actual `update()` function rather than replacing the existing structure. Replaced the list-comprehension removal with explicit miss detection so each missed balloon decrements lives.

USER

Task 4.

CHATGPT

Implemented a 30-second round using `time.time()`, stopped spawning and accepting clicks after game over, displayed the remaining time, displayed the final score, and added a `restart()` method that resets balloons, score, lives, `game_over`, and the round timer.

USER

Shared `game_engine.py` and `main.py`.

CHATGPT

Reviewed the actual files and identified that `main.py` needed `KEYDOWN` handling for the R key. Added the restart trigger only when `engine.game_over` is true.

USER

Pressing R caused `AttributeError: GameEngine has no attribute restart`.

CHATGPT

Diagnosed the issue as the `restart()` method not being present correctly inside the `GameEngine` class. Provided the correctly indented `restart()` method and explained that the `pygame pkg_resources` message was only a warning, not the crash cause.

USER

Shared the corrected `game_engine.py`.

CHATGPT

Verified that `restart()` was correctly indented inside `GameEngine` and that Task 4's timer, lives, game-over, and restart logic were integrated correctly.

CHATGPT

Final intended test sequence: verify Time starts at 30 and counts down, verify losing three lives ends the round, verify timer expiry ends the round, verify clicks are ignored after game over, and verify pressing R resets score to 0, lives to 3, timer to 30, and starts a new round.

Final Git Commit Structure

Task-4: Add timed rounds and restart

Task-3: Add lives and miss system

Task-2: Add different balloon types

Task-1: Fix balloon click detection

Initial Balloon Pop starter

Assignment Completion Notes

The conversation followed the lab workflow: inspect the README, run the starter game, use ChatGPT as the debugging/pair-programming partner, implement the four tasks individually, maintain separate commits, and prepare the project for the final after-change gameplay recording. The before and after videos and the final chat-history submission remain the student's responsibility.